USB-C, ESD

J1 USB-C

VBUS A4 · CC1 A5 · CC2 B5 · D- B7 · D- A7 · D+ A6 · D+ B6 · SBU1 A8 · SBU2 B8 · GND A1 · SHIELD · S1

VBUS, PWR_FLAG

C1 100n

USB_DM_C, USB_DP_C → U2 USBLC6-2SC6 → USB_DM, USB_DP

R1 5.1k, R2 5.1k

GND, PWR_FLAG

Shield straight to GND

Rd: USB-C sink

3V3 LDO

U3 AP7361C-33FGE

VIN 8 · EN 5 · GND 4 · VOUT 1

C2 2.2u, C3 10u

+3V3

277 mA max, 0.62 W at 5.5 V in: EP to GND vias

RP2350A, FLASH, CRYSTAL

100 nF at IOVDD pins 1, 11, 20, 30, 38 (C4–C8), ADC_AVDD 44 and IOVDD 45 (C9, shared as in the vendor layout), USB_OTP_VDD 53 and QSPI_IOVDD 54 (C11). The Decouples field on each names its pin.

Regulator: C14 CIN and C15 COUT (0402 small pads). L1's polarity and copper all follow chips/rp2350.md

C15 COUT at L1; C16 at DVDD pin 23; C17, C18 at DVDD pins 6, 39

L1 3.3u

PWR_FLAG +1V1

C15 4.7u, C16 4.7u, C17 100n, C18 100n

C4 100n, C5 100n, C6 100n, C7 100n, C8 100n, C11 100n, C9 100n

R3 33R

C13 4.7u, C14 4.7u

U1 RP2350A

VREG_AVDD 46 · USB_OTP_VDD 53 · ADC_AVDD 44 · QSPI_IOVDD 54 · IOVDD 1 · VREG_VIN 49 · VREG_LX 48 · VREG_FB 50 · DVDD 6

RUN 26

27R at the chip

R4 27R USB_DM → USB_DM_MCU 51 · USB_DM

R5 27R USB_DP → USB_DP_MCU 52 · USB_DP

+3V3, C22 100n

U5 W25Q128JVSIQ

QSPI_SS 60 · QSPI_SCLK 56 · QSPI_SD0 57 · QSPI_SD1 59 · QSPI_SD2 58 · QSPI_SD3 55

Y1 12MHz, C20 15p, C21 15p, R6 1k

XIN 21 · XOUT 22

SWCLK 24 · SWDIO 25

VREG_PGND, GND 47, 61

GPIO0 2 · GPIO1 3 · GPIO2 4 · GPIO3 5 · GPIO4 7 · GPIO5 8 · GPIO6 9 · GPIO7 10 · GPIO8 12 · GPIO9 13 · GPIO10 14 · GPIO11 15 · GPIO12 16 · GPIO13 17 · GPIO14 18 · GPIO15 19 · GPIO16 27 · GPIO17 28 · GPIO18 29 · GPIO19 31 · GPIO20 32 · GPIO21 33 · GPIO22 34 · GPIO23 35 · GPIO24 36 · GPIO25 37 · GPIO26/ADC0 40 · GPIO27/ADC1 41 · GPIO28/ADC2 42 · GPIO29/ADC3 43

VREG_PGND joins GND at one point

GPIO HEADERS (TOP VIEW)

USB-C end

J2: GPIO0 1 · GPIO1 2 · 3 · GPIO2 4 · GPIO3 5 · GPIO4 6 · GPIO5 7 · 8 · GPIO6 9 · GPIO7 10 · GPIO8 11 · GPIO9 12 · 13 · GPIO10 14 · GPIO11 15 · GPIO12 16 · GPIO13 17 · 18 · GPIO14 19 · GPIO15 20 — GPIO left

J3: 20 VBUS · 19 GPIO29 · 18 · 17 GPIO28 · 16 +3V3 · 15 GPIO27 · 14 GPIO26 · 13 GPIO25 · 12 GPIO24 · 11 GPIO23 · 10 RUN · 9 GPIO22 · 8 · 7 GPIO21 · 6 GPIO20 · 5 GPIO19 · 4 GPIO18 · 3 · 2 GPIO17 · 1 GPIO16 — GPIO right

Pin map: parts/user-io.md. Not Pico pin-compatible: pins 31–39 differ. Headers are not assembled.

SWD, STATUS LED

J4 SWD: SWCLK 1 · GND 2 · SWDIO 3

R9 1k, D1 RED, GPIO25

RGB LEDS: VLED RAIL, LEVEL SHIFTER, 4 PIXELS

VBUS, D2 SM4007PL, R11 4.7k, D7 BZT52B5V1

VLED, C23 4.7u, C24 100n

PWR_FLAG

U4 SN74LV1T125DBVR

GPIO23, R12 100k, R10 33R, LED_DIN

C25 100n, C26 100n, C27 100n, C28 100n

D3, D4, D5, D6 WS2812B-2020-V6

PIX12, PIX23, PIX34

VLED 3.5–5.1 V: the rectifier's drop, held up by the 4.7k bleed, keeps the pixels inside 3.3–5.3 V; D7 clamps a hot-plug ring (parts/rgb-leds.md)

BUTTONS

R7 1k, QSPI_SS → SW1 BOOTSEL

R8 1k, RUN → SW2 RESET

Drawn by scripts/schematic.py	
Author: Sequoia Alexander	
CERN-OHL-P-2.0	
RP2350A dev board, 21 x 51 mm	
github.com/sequoia-hope/easy_pcb	
Sheet: /	
File: easy2350.kicad_sch	
Title: easy2350	
Size: A3	Date: 2026-10-23 — **Rev: A**
KiCad E.D.A. 9.0.8	Id: 1/1